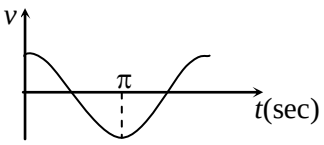
# PAPER-II
# PHYSICS
## SECTION – I
### Single Correct Choice Type

**1.** Length of the plate = 8.32 – (– 0.02) cm = 8.34 cm

$\therefore$ **(b)**

**2.** $x = \sin \omega t$

$v = \dfrac{dx}{dt} = \cos \omega t$




Since speed of particle decreases during

$(t = 0 \sec \text{ to } t = \dfrac{\pi}{2} \sec)$ and $(t = \pi \sec \text{ to } t = \dfrac{3\pi}{2} \sec)$

$\therefore$ **(b)**

**3.** When system will again come to rest the block will move $(l + 3l + l = 5l)$ distance on plank.

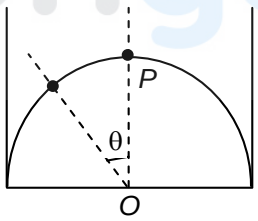
Let maximum displacement of plank be $\Delta x$

Since centre of mass remain at rest

$$\Delta x = \frac{5ml}{m + M}$$

$\therefore$ **(a)**

**4.** When the bead will reach at *P*, there will be maximum contact force between bead and structure. If the mass of bead is m and its speed at *P* is *v* then




Net work done by buoyancy and gravity = Δ K.E.

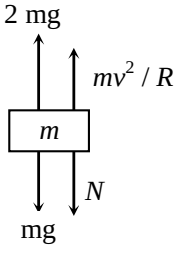
$$\text{mg } R(1 - \cos\theta) = \frac{1}{2}mv^2$$

$$mgR\left(2\sin^2\frac{\theta}{2}\right) = \frac{1}{2}mv^2$$

$$mgR\,2\frac{\theta^2}{4} = \frac{1}{2}mv^2$$

$$\Rightarrow \quad \frac{mv^2}{R} = mg\theta^2$$

Drawing *FBD* of bead at *P*.




$$N + mg = 2mg + \frac{mv^2}{R}$$

$$N = mg + \frac{mv^2}{R} = mg + mg\theta^2 = mg(1 + \theta^2)$$

$$N \,\alpha\, (1 + \theta^2)$$

$\therefore$ **(b)**

**5.** $\Delta Q = \Delta U + \Delta W$

$CdT = C_v dT + PdV$

$C = C_v + P\dfrac{dV}{dT}$ …(1)

Since $PV = RT$

$P\dfrac{dV}{dT} + V\dfrac{dp}{dT} = R$ …(2)

and $T^2 = 4P$

$2T = 4\dfrac{dP}{dT} \Rightarrow \dfrac{dP}{dT} = \dfrac{T}{2}$ …(3)

from (2) and (3) $P\dfrac{dV}{dT} + \dfrac{VT}{2} = R$

$P\dfrac{dV}{dT} + 2R = R$

$P\dfrac{dV}{dT} = -R$ …(4)

From (1) and (4) $C = C_v - R = \dfrac{3}{2}R - R = \dfrac{R}{2}$

$\therefore$ **(c)**

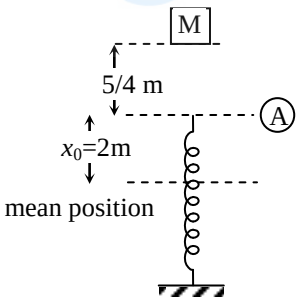
**6.** Say the block takes time $t_1$ to reach at $A$. $\dfrac{1}{2}gt_1^2 = \dfrac{5}{4} \Rightarrow t_1 = \dfrac{1}{2}$ sec

Now block will perform SHM.

Let at time of force balanced position compression in spring is $x_0$ then

$kx_0 = mg \Rightarrow 5x_0 = 10 \Rightarrow x_0 = 2\,m$

This will be the mean position of SHM. Maximum compression in the spring is if $y_0$ then

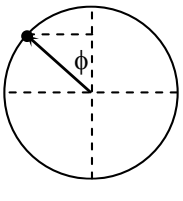



$mg\left(\dfrac{5}{4} + y_0\right) = \dfrac{1}{2}ky_0^2 \Rightarrow 10\left(\dfrac{5}{4} + y_0\right) = \dfrac{5}{2}y_0^2 \Rightarrow \quad y_0 = 5\text{m}$

Therefore block will perform SHM with amplitude $A = (5-2)\,\text{m} = 3\text{m}.$ with angular frequency $\omega = \sqrt{\dfrac{k}{m}} = \sqrt{5}$ mean position is shown. Let block performs SHM for time $T$.

Drawing phasor of SHM.

$3\cos\phi = 2 \Rightarrow \cos\phi = \dfrac{2}{3} \Rightarrow \phi = \cos^{-1}\left(\dfrac{2}{3}\right)$




So $T = \dfrac{2\pi - 2\phi}{\omega} = \dfrac{2\pi}{\sqrt{5}} - \dfrac{2}{\sqrt{5}}\cos^{-1}\left(\dfrac{2}{3}\right)$

So overall time it takes to gain its initial position again $= 2t_1 + T$

$= 1 + \frac{2\pi}{\sqrt{5}} - \frac{2}{\sqrt{5}}\cos^{-1}\left(\frac{2}{3}\right)$

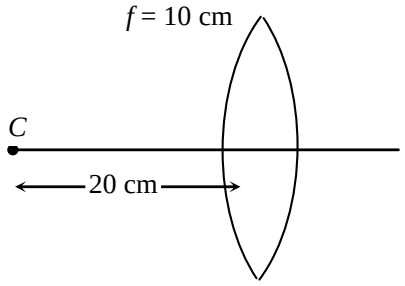
$\therefore$ **(b)**

**7.** **Case – 1**

$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$\frac{1}{v} - \frac{1}{(-20)} = \frac{1}{10} \Rightarrow \frac{1}{v} = \frac{2}{20} - \frac{1}{20} \Rightarrow v = 20\,\text{cm}$

Magnification $= \frac{v}{u} = \frac{20}{20} = -1 = m_1$

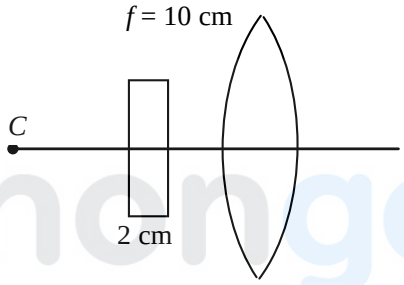



**Case – 2**

$u = -\left(18 + \frac{2}{\mu}\right)$

$\frac{1}{v} - \frac{1}{-\left(18 + \frac{2}{\mu}\right)} = \frac{1}{10}$

$\Rightarrow \frac{1}{v} = \frac{1}{10} - \frac{1}{\left(18 + \frac{2}{\mu}\right)}$



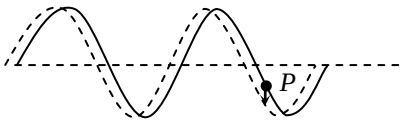

$\Rightarrow \frac{1}{v} = \frac{18 + \frac{2}{\mu} - 10}{10\left(18 + \frac{2}{\mu}\right)} = \frac{8 + \frac{2}{\mu}}{10\left(18 + \frac{2}{\mu}\right)}\,\text{cm.}$

Magnification $m_2 = \frac{\frac{10\left(18 + \frac{2}{\mu}\right)}{\left(8 + \frac{2}{\mu}\right)}}{-\left(18 + \frac{2}{\mu}\right)} = -\frac{1}{2} = \frac{m_1}{2}$

$\Rightarrow \frac{10}{8 + \frac{2}{\mu}} = \frac{1}{2} \Rightarrow 20 = 8 + \frac{2}{\mu} \Rightarrow \frac{2}{\mu} = 12 \Rightarrow \mu = \frac{1}{6} = 0.166 = 0.17$

$\therefore$ **(b)**

**8.** Drawing the shape of string after sometime.




Its clear that velocity of $P$ at this instant is downwards.

$\therefore$ **(b)**

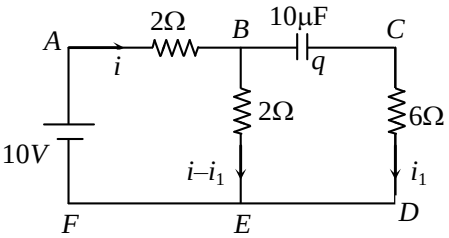
## SECTION – II
### Multiple Correct Choice Type

**9.** for loop $ABEFA$

$10 = 2i + 2(i - i_1) \Rightarrow 5 = 2i - i_1$ …(1)

for loop $BCDEB$

$2(i - i_1) = \frac{q}{C} + 6i_1 \Rightarrow 2i = \frac{q}{C} + 8i_1$ …(2)




from (1) and (2)

$5 + i_1 = \frac{q}{C} + 8i_1 \Rightarrow 7i_1 + \frac{q}{C} = 5$ since $i_i = \frac{dq}{dt}$

or $7\frac{dq}{dt} + \frac{q}{C} = 5$

$7\frac{dq}{dt} = \frac{5C - q}{C}$

$\int_0^q \frac{dq}{5C - q} = \int_0^q \frac{dt}{7C} \Rightarrow -\left[\ln(5C - q)\right]_0^q = \frac{t}{7C}$

$\Rightarrow \frac{5C - q}{5C} = e^{-\frac{t}{7C}}$

$q = 5C\left(1 - e^{-t/7C}\right) = q_0\left(1 - e^{-t/7C}\right)$

Time constant $\tau = 7C = 7 \times 10 \times 10^{-6}$ sec $= 70 \times 10^{-6}$ sec. $= 70\mu$sec.

In steady state charge on capacitor $q = q_0 = 5C = 50\mu C$ at any instant current through $6\Omega$ is $i_1 = \frac{5}{7}e^{-t/\tau}, \tau = 70\mu$sec.

$\therefore$ **(a, b, c)**

**10.** $N = mg + qE$ but $\frac{q}{m} = 8 \Rightarrow q = 8m$

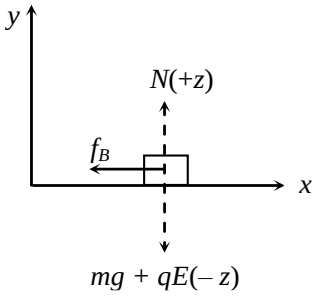
$N = mg + 8mE$

$= 50m$

$f = \mu N = \frac{50}{3}m$

$a_t = \frac{f}{m} = -\frac{50}{3}$ m/s

Since $v = u + a_t t$



$$0 = 100 - \frac{50}{3}t \Rightarrow t = 6\,\text{sec}$$

$$s = ut + \frac{1}{2}a_t t^2$$

$$s = 100(6) - \frac{1}{2}\left(\frac{50}{3}\right)(6)^2 = s = 300\ m$$

Since force due to electric and magnetic field on the particle is perpendicular to the displacement so work done by electric field and magnetic field is zero.

∴ **(a, b, c, d)**

**11.** For the considered region given charge distribution will act as a dipole system. Therefore electric field anywhere in the region is directed towards +ve *x*-axis.

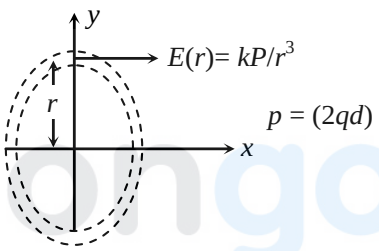
Due to symmetry we can conclude that electric potential anywhere in *y*-*z* plane will be zero.

So the work done from bringing a positive test change from $\left(0, \frac{a}{\sqrt{2}}, \frac{a}{\sqrt{2}}\right)$ to $\left(a, \frac{a}{\sqrt{2}}, \frac{a}{\sqrt{2}}\right)$ is zero.

Electric flux crossing

$$\phi = \int_{r=a}^{b} E(r)(2\pi r dr) = (kP)(2\pi)\int_a^b \frac{dr}{r^2}$$

$$= \frac{1}{4\pi \in_0}(2qd)(2\pi)\left(\frac{-1}{r}\right)_a^b = \frac{qd}{\in_0}\left(\frac{1}{a} - \frac{1}{b}\right)$$




∴ **(a, b, c, d)**

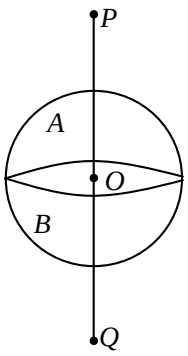
**12.** Lets assume the complete sphere. Upper hemisphere is denoted by *A* and lower by *B*.

Combined field intensity at *P* or *Q* is

$$E = \frac{GM}{(2R)^2} = \frac{G}{4R^2}\frac{4}{3}\pi R^3 \rho = \frac{\pi G \rho R}{3}$$

Its obvious that intensity at *P* and *Q* will be unlike parallel.

Its clear that if magnitude of intensity at *Q* due to only the hemisphere *B* is $E_0$ then at *P* it will be $\frac{\pi}{3}G\rho R - E_0$




∴ **(b, d)**

## SECTION–III
## Comprehension Type

**13.** $\frac{1}{3}$ mole of $O_2$ dissociates into atoms.

$\frac{2}{3}$ mole of O formed due to dissociation.

∴ No. of moles of monoatomic gas in mixture = $1 + \frac{2}{3} = \frac{5}{3}$ mole

$\therefore$ No. of moles of diatomic gas in mixture = $\frac{2}{3}$ mole

$$\frac{\mu_{mix}}{\gamma_{mix-1}} = \frac{\frac{5}{3}}{\frac{5}{3}-1} + \frac{\frac{2}{3}}{\frac{7}{5}-1}$$

$\gamma_{mix} = 1.56$

$\therefore$ **(a)**

**14.** For backward motion of piston,

$$TV^{\gamma-1} = T_0V_0^{\gamma-1}$$

$$T = T_0\left(\frac{V_0}{V_0/2}\right)^{1.56-1} = \sqrt{2}\,T_0$$

$\therefore$ **(a)**

**15.** $\Delta U = \mu_{mix}\frac{R}{\gamma_{mix-1}}(T-T_0) = \frac{25}{6}R\left(\sqrt{2}-1\right)T_0 = 14.18\ T_0$

$\therefore$ **(a)**

## SECTION–IV
## Matrix–Match Type

**1.** **A – (4, 5); B – (1, 2, 3); (C) – 1, 2, 5; (D) – (3, 4)**

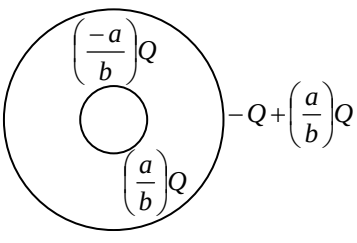
**2.** **A – (1, 4); B – (3, 4); C – (1, 5); D – (2, 4)**

## SECTION–V
## Integer Answer Type

**1.** When outer surface is grounded charge '$-Q$' resides on the inner surface of sphere '$B$'

Now sphere $A$ is connected to earth potential on its surface becomes zero.

Let the charge on the surface $A$ becomes $q$




$$\frac{kq}{a} - \frac{kQ}{b} = 0$$

$$\Rightarrow \quad q = \frac{a}{b}Q$$

Consider the figure. In this position energy stored

$$E_1 = \frac{1}{8\pi\varepsilon_0 a}\left[\frac{a}{b}Q\right]^2 + \frac{Q^2}{8\pi\varepsilon_0 b} + \frac{1}{4\pi\varepsilon_0 b}\left[\frac{a}{b}Q\right](-Q)$$

when '$S_3$' is closed, total charge will appear on the outer surface of shell '$B$'. In this position energy stored $E_2 = \frac{1}{8\pi\varepsilon_0 b}\left(\frac{a}{b}-1\right)^2 Q^2$

Heat produced = $E_1 - E_2 = \dfrac{Q^2 a(b-a)}{8\pi\varepsilon_0 b^3} = 1.8$

$\therefore$ **9.**

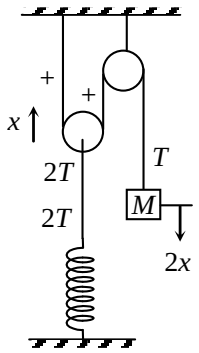
**Ans. 9**

**2.** Let elongation in the spring is $x_0$ when $M$ is in equilibrium

$T = Mg$

and $kx_0 = 2T = 2Mg$

$x_0 = \dfrac{2Mg}{k}$




So maximum amplitude = $2x_0 = \dfrac{4Mg}{k}$

$\therefore \quad \alpha = 4$

**Ans. 4**

**3.** $F_{net} = F_1 - F_2$

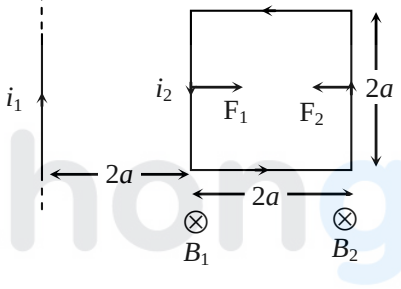
$= i_2 B_1(2a) - i_2(B_2)(2a)$

$= \left(\dfrac{\mu_0}{4\pi}\dfrac{2i_1}{2a}\right) i_2(2a) - \left(\dfrac{\mu_0}{4x}\dfrac{2i_1}{4a}\right) i_2(2a)$

$= \dfrac{2\mu_0}{4\pi} - \dfrac{\mu_0}{4\pi} = \dfrac{\mu_0}{4\pi}$

$\therefore \quad \beta = 1$

**Ans. 1**




**4.** Resistance of potentiometer wire $R' = \int_0^1 \lambda dx = 2\int_0^1 x dx = 1\Omega$

$\Rightarrow \quad E_S = \left(\dfrac{E_p}{R'+R}\right).\int_0^{\frac{1}{2}} \lambda dx = \left(\dfrac{E_p}{1+t^2}\right)\dfrac{1}{4}$

$E_S = \left(\dfrac{E_P}{1+1}\right)\dfrac{1}{4} = \dfrac{E_P}{8}$

$\therefore \quad \gamma = 8$

**Ans. 8**